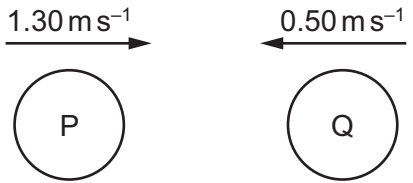
8 Two balls P and Q, of equal mass, move along a straight line directly towards each other as shown.

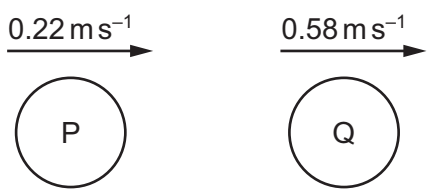


Ball P has velocity $1.30\,m\,s^{-1}$ to the right. Ball Q has velocity $0.50\,m\,s^{-1}$ to the left.

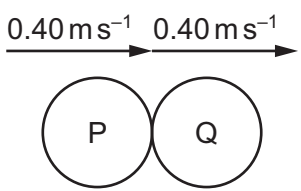
P and Q collide with one another. The collision is perfectly elastic and the total momentum is conserved.

Which diagram correctly shows the motion of P and Q after the collision?

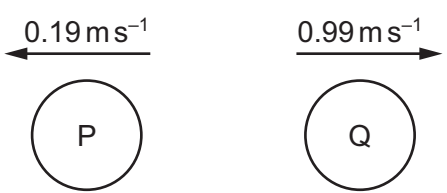
A



B



C



D

